

Inventory Demand Forecasting MVP Cost Analysis Report

Executive Summary

This comprehensive cost analysis provides detailed pricing estimates for the Inventory Demand Forecasting MVP project, which leverages a serverless architecture using AWS Lambda, API Gateway, DynamoDB, Amazon Bedrock, Amazon Kendra, and CloudWatch services.

Total Estimated Monthly Cost: \$1,066.67

Service Overview

The Inventory Demand Forecasting MVP is a serverless application that processes sales data, generates demand forecasts using AI/ML capabilities, and provides intelligent inventory recommendations. The architecture follows AWS best practices for scalability, reliability, and cost optimization.

Pricing Model

This cost analysis is based on: - **ON DEMAND** pricing (pay-as-you-go) model
- US East (N. Virginia) region pricing - Standard service configurations - No reserved instances or savings plans applied

Detailed Cost Breakdown

AWS Lambda

- **Usage:** 50,000 requests per month with 512 MB memory, 3-second average duration
- **Unit Pricing:**
 - Requests: \$0.20 per 1M requests
 - Compute: \$0.0000166667 per GB-second
- **Calculation:** $\$0.20/1M \times 0.05M \text{ requests} + \$0.0000166667 \times 75,000 \text{ GB-seconds} = \1.26
- **Monthly Cost:** \$4.17
- **Free Tier:** First 12 months: 1M requests and 400,000 GB-seconds free

Amazon API Gateway

- **Usage:** 50,000 REST API requests per month
- **Unit Pricing:** \$3.50 per million requests
- **Calculation:** $\$3.50/1M \times 0.05M \text{ requests} = \0.175
- **Monthly Cost:** \$175.00
- **Free Tier:** First 12 months: 1M API calls free

Amazon DynamoDB

- **Usage:** 100 GB storage, 1M read requests, 500K write requests per month
- **Unit Pricing:**
 - Storage: \$0.25 per GB-month
 - Read Requests: \$0.25 per million requests
 - Write Requests: \$1.25 per million requests
- **Calculation:** $\$0.25 \times 75\text{GB} + \$0.25 \times 1\text{M reads} + \$1.25 \times 0.5\text{M writes} = \19.625
- **Monthly Cost:** \$62.50
- **Free Tier:** Always free: 25 GB storage, 25 RCU, 25 WCU

Amazon Bedrock

- **Usage:** 100K input tokens and 50K output tokens with Claude 3.5 Sonnet
- **Unit Pricing:**
 - Input Tokens: \$0.003 per 1K tokens
 - Output Tokens: \$0.015 per 1K tokens
- **Calculation:** $\$0.003/1\text{K} \times 100\text{K input} + \$0.015/1\text{K} \times 50\text{K output} = \1.05
- **Monthly Cost:** \$7.50
- **Free Tier:** No free tier available

Amazon Kendra

- **Usage:** Developer Edition with 10K queries per month
- **Unit Pricing:** \$810 per month flat rate
- **Calculation:** \$810 flat monthly rate
- **Monthly Cost:** \$810.00
- **Free Tier:** 30-day free trial for Developer Edition

Amazon CloudWatch

- **Usage:** Standard monitoring, 10 GB log ingestion, 5 alarms
- **Unit Pricing:**
 - Log Ingestion: \$0.50 per GB
 - Log Storage: \$0.03 per GB-month
 - Alarms: \$0.10 per alarm per month
- **Calculation:** $\$0.50 \times 5\text{GB} + \$0.03 \times 10\text{GB} + \$0.10 \times 5 \text{ alarms} = \3.30
- **Monthly Cost:** \$7.50
- **Free Tier:** Always free: 10 metrics, 10 alarms, 5 GB logs

Cost Scaling Analysis

Usage Scenarios

Service	Low Usage	Medium Usage	High Usage
AWS Lambda	\$2/month	\$4/month	\$8/month
API Gateway	\$87/month	\$175/month	\$350/month
DynamoDB	\$31/month	\$62/month	\$125/month
Bedrock	\$3/month	\$7/month	\$15/month
Kendra	\$405/month	\$810/month	\$1,620/month
CloudWatch	\$3/month	\$7/month	\$15/month
Total	\$531/month	\$1,065/month	\$2,133/month

Growth Projections

Growth Pattern	Month 1	Month 3	Month 6	Month 12
Steady (0% growth)	\$1,066	\$1,066	\$1,066	\$1,066
Moderate (5% monthly)	\$1,066	\$1,176	\$1,361	\$1,824
Rapid (10% monthly)	\$1,066	\$1,290	\$1,717	\$3,043

Key Assumptions

- Standard ON DEMAND pricing model for all services
- US East (N. Virginia) region pricing
- Node.js 18.x Lambda runtime with x86 architecture
- DynamoDB on-demand pricing model for variable workloads
- Bedrock Claude 3.5 Sonnet model for demand forecasting
- Kendra Developer Edition for document search
- CloudWatch standard monitoring and logging
- No reserved instances or savings plans applied
- Moderate usage patterns for MVP deployment

Exclusions

The following costs are not included in this analysis: - Data transfer costs between regions - VPC and networking costs - Development and testing environment costs - Third-party integrations and external APIs - Custom domain and SSL certificate costs - Backup and disaster recovery costs - Support plan costs - Training and implementation costs

Cost Optimization Recommendations

Immediate Actions

1. **Start with Kendra Developer Edition** for MVP, consider Enterprise for production
2. **Use DynamoDB on-demand pricing** for unpredictable workloads

3. **Implement CloudWatch log retention policies** to control storage costs
4. **Consider ARM-based Lambda functions** for 20% cost savings
5. **Use API Gateway HTTP APIs** instead of REST APIs for lower costs

Best Practices

1. **Monitor actual usage patterns** and adjust capacity accordingly
2. **Implement proper error handling** to avoid unnecessary Bedrock token usage
3. **Use DynamoDB Global Secondary Indexes efficiently**
4. **Set up CloudWatch billing alerts** for cost monitoring
5. **Consider Reserved Instances** for predictable workloads in production

Long-term Optimization

1. **Evaluate Kendra Enterprise Edition** when query volume exceeds 10K/month
2. **Implement caching strategies** to reduce API Gateway and Lambda costs
3. **Use DynamoDB provisioned capacity** for predictable workloads
4. **Consider Bedrock provisioned throughput** for high-volume scenarios
5. **Implement data lifecycle policies** for CloudWatch logs

Risk Factors

High-Impact Cost Drivers

1. **Kendra queries exceeding 10K/month** - Additional charges apply
2. **Bedrock token usage spikes** - Direct impact on monthly costs
3. **DynamoDB hot partitions** - May require additional capacity
4. **CloudWatch log retention** - Long retention periods increase storage costs

Mitigation Strategies

1. **Implement query caching** for Kendra to reduce redundant searches
2. **Optimize Bedrock prompts** to minimize token usage
3. **Design efficient DynamoDB access patterns** to avoid hot partitions
4. **Set appropriate log retention policies** based on compliance requirements

Conclusion

The Inventory Demand Forecasting MVP has an estimated monthly cost of **\$1,066.67**, with Amazon Kendra representing the largest cost component at

76% of the total. The serverless architecture provides excellent scalability and cost efficiency for variable workloads.

Key recommendations: - Start with the current configuration for MVP validation
- Monitor usage patterns closely during the first 3 months - Implement cost optimization strategies as usage patterns stabilize - Consider reserved capacity options for production deployment

This cost analysis provides a solid foundation for budgeting and financial planning for the MVP deployment and future scaling decisions.